

Mathematics for Computer Science

COMP1021 - Durham University

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Part I

Calculus

1 Introduction to Calculus

1.1 Sets

Definition 1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 1.2.1:** A is known as the domain of f .
- **Definition 1.2.2:** B is known as the codomain of f .
- **Definition 1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

1.3 Hyperbolic Functions

Definition 1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

2 Multivariate Functions

2.1 Multivariate Derivatives

Definition 2.1.1: For a function $f(x, y)$, if we take $f = f_0$ to be fixed but unknown, then we can differentiate f as a function of x at $x = x_0$, giving us the partial derivative of f with respect to x which gives the gradient when moving in the x -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left(\frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (3)$$

Definition 2.1.2: The vector of partial derivatives, denoted as ∇f , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of f , where its magnitude is the rate of increase in that direction. For example, for a function $f(x, y)$,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (4)$$

Definition 2.1.3: The directional derivative of the function $f(x, y)$ gives the gradient at any point whilst looking in the direction given by the unit vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$.

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (5)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (6)$$

From this, we get that, for a function $f(\mathbf{x}_0)$ and a direction $\mathbf{v}$,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (7)$$

2.2 Multivariate Extrema

Definition 2.2.1: A function that is continuous is in class C^0 , and a function that is continuous and has a continuous derivative is in class C^1 . A function f such that f' , f'' , ..., $f^{(k)}$ exist and are continuous is in class C^k .

Definition 2.2.2: A function f such that derivatives of all orders exist and are continuous is in class C^∞ is also smooth.

Definition 2.2.3: All the second-order partial derivatives of a function $f(x, y)$ can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (8)$$

This can be used to determine the form of bivariate extrema. Suppose $f(x, y) \in C^2$ and $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ then,

- (x_0, y_0) is a local maximum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} < 0$ at (x_0, y_0) .
- (x_0, y_0) is a local minimum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} > 0$ at (x_0, y_0) .
- (x_0, y_0) is a saddle point if $\det(\mathbf{H}_f) < 0$.
- If $\det(\mathbf{H}_f) = 0$, then the test is inconclusive.

At a stationary point x_0 , the quadratic approximation of a function $f(x)$ is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (9)$$

But as $\mathbf{H}_f(x_0)$ is real and symmetric, it has orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (10)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (11)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point x_0 ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

2.3 Vector-Valued Functions

Definition 2.3.1: A vector-valued function takes an n -dimensional vector as input, and outputs an m -dimensional vector, e.g., $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition 2.3.2: The Jacobian matrix of a vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be formed by setting $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$ to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (12)$$

For a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix can be used to linearly approximate f at x ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (13)$$

3 Lagrange Multipliers

3.1 Constrained Optimisation

Definition 3.1.1: The goal of a constrained optimisation problem is to either minimise or maximise a multivariate function $f(\mathbf{x})$ where $x_1, x_2, \dots, x_n$ are not independent subject to some constraint $g(\mathbf{x})$.

Definition 3.1.2: The Lagrange multipliers method is used to find stationary points of an objective function $f(\mathbf{x}) \in C^1$ subject to the constraint $g(\mathbf{x}) = 0$.

1. Define a new objective function called the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x}) \quad (14)$$

2. Find the stationary points of $\mathcal{L}$ with respect to both $\mathbf{x}$ and λ , i.e., solve the system of equations $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$ and $g(\mathbf{x}) = 0$ for the unknowns $\mathbf{x}$ and λ .
3. Then, the constrained extrema are found among the solutions to these equations. Identify their nature.

4 Series

4.1 Introduction to Series

Definition 4.1.1: A sequence is an ordered, countably infinite, set of numbers.

Definition 4.1.2: A series is the sum of the members of a sequence.

Theorem 4.1.1: If the series $\sum_{n=0}^{\infty} a_n$ and the series $\sum_{n=0}^{\infty} b_n$ are convergent, and have respective sums α and β , then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta \quad (15)$$

and

$$\sum_{n=0}^{\infty} (a_n - b_n) = \alpha - \beta \quad (16)$$

Proof: $\sum_{n=0}^{\infty} a_n$ converges to α and $\sum_{n=0}^{\infty} b_n$ converges to β . Given an arbitrary $\epsilon > 0$, let $\epsilon_a = \frac{1}{2}\epsilon = \epsilon_b$. Since $\sum_{n=0}^{\infty} a_n$ converges, $\exists N_a$ such that,

$$\left| \sum_{r=0}^n a_r - \alpha \right| < \epsilon_a, \quad \forall n > N_a \quad (17)$$

Similarly, since $\sum_{n=0}^{\infty} b_n$ converges, $\exists N_b$ such that,

$$\left| \sum_{r=0}^n b_r - \beta \right| < \epsilon_b, \quad \forall n > N_b \quad (18)$$

Taking $N = \max\{N_a, N_b\}$, then $\forall n > N$,

$$\begin{aligned} \left| \sum_{r=0}^n (a_r + b_r) - (\alpha + \beta) \right| &= \left| \left(\sum_{r=0}^n a_r - \alpha \right) + \left(\sum_{r=0}^n b_r - \beta \right) \right| \\ &\leq \left| \sum_{r=0}^n a_r - \alpha \right| + \left| \sum_{r=0}^n b_r - \beta \right| \leq \epsilon_a + \epsilon_b = \epsilon \end{aligned} \quad (19)$$

i.e, $\sum_{n=0}^{\infty} (a_n + b_n)$ converges to $(\alpha + \beta)$ using triangle inequality $|a + b| \leq |a| + |b|$.

□

Theorem 4.1.2: If $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=0}^{\infty} a_n$ diverges.

Proof: Suppose $\sum_{n=0}^{\infty} a_n$ converges for some S , Then, $\lim_{n \rightarrow \infty} S_n = S$, which means that given arbitrary $\epsilon > 0$, there exists some N_ϵ such that, for all $n > N_\epsilon$,

$$|S_n - S| < \epsilon \quad |S_n - S| < 2\epsilon \quad (20)$$

Combining these,

$$2\epsilon > |S_n - S| + |S_{n+1} - S| = |S - S_n| + |S_{n+1} - S| \geq |S_{n+1} - S_n| = |a_{n+1}| \quad (21)$$

But ϵ was arbitrary, so we have shown that, for any $\epsilon' > 0$ there exists an N_ϵ such that $|a_n| < \epsilon'$ for all $n > N_\epsilon$.

□

As the number of elements in a series approaches infinity...

- **Definition 4.1.3:** If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.
- **Definition 4.1.4:** If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.
- **Definition 4.1.5:** If $\lim_{n \rightarrow \infty} |a_n| = 0$, then the series absolutely converges.
- **Definition 4.1.6:** If $\lim_{n \rightarrow \infty} a_n = S$, then the series is convergent to S .

Theorem 4.1.3: The ratio test states that, if $\sum_{n=1}^{\infty} a_n$ is a series where $a_n \neq 0$ and $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ then

- If $L < 1$, the series absolutely converges.
- If $L > 1$, the series is divergent.
- If $L = 1$, then the test fails.

Proof: (a) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then $\exists r$ with $L < r < 1$ and N such that $\forall n > N$:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < r \quad \rightarrow \quad |a_{n+1}| < r|a_n| \quad (22)$$

Then $|a_{n+2}| < r|a_{n+1}| < r^2|a_n|$ and $|a_{n+3}| < r|a_{n+2}| < r^3|a_n|$ etc. So the remainder is

$$R_{n+m} = \lim_{m \rightarrow \infty} \left(|a_n| r^{m+1} \frac{1}{1-r} \right) = 0 \quad (23)$$

Hence the series converges.

□

Definition 4.1.7: The n^{th} root test states that, for the series $\sum_{n=1}^{\infty} a_n$, $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$,

- If $L < 1$, then the series absolutely converges.
- If $L > 1$, then the series is divergent.
- If $L = 1$, then the test fails.

Definition 4.1.8: The alternating series test states that the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges if $a_n > 0$, $a_{n+1} \leq a_n \forall n$, and $\lim_{n \rightarrow \infty} a_n = 0$

Theorem 4.1.4: Every subsequence of a convergent sequence converges to the same limit as the original sequence.

Proof: Let $\{a_{n_k}\}$ be a subsequence of $\{a_n\}$. First note that $n_k \geq k$ and so $n_{k+1} \geq k+1$. Let $\lim_{n \rightarrow \infty} a_n = L$. Then, for any $\epsilon > 0$, there exists N_ϵ such that for all $n > N_\epsilon$ it holds that $|a_n - L| < \epsilon$. Let $k > N_\epsilon$, then $n_k > N_\epsilon$ and so it must be that $|a_{n_k} - L| < \epsilon$.

□

4.2 Power Series

Definition 4.2.1: A power series is a series involving a variable x and a constant x_0 of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \quad (24)$$

At any specific value of x , the power series becomes a normal series.

Definition 4.2.2: For an arbitrary power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$, we can use the ratio test (including the $(x - x_0)$ term). From this, we call $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$ the radius of convergence. If $|x - x_0| < r$, we have absolute convergence and if $|x - x_0| > r$, we have divergence.

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $r > 0$. Then, within the interval $(-r, r)$:

- $f(x)$ is continuous.
- $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ (term-by-term differentiation).
- $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ (term-by-term integration).
- Both series $f'(x)$ and $\int_0^x f(t) dt$ have the same radius of convergence as $f(x)$.

Definition 4.2.3: Suppose we start with some function $f(x)$ and we wish to determine a power series $\sum_{n=0}^{\infty} a_n x^n$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Putting these values in, we see that if such a power series exists, then:

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f^{(3)}(0)\frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} \quad (25)$$

This is called the Maclaurin series for f . To use the series, we must have that $f \in C^\infty$ and the power series converges.

Definition 4.2.4: We can recenter the power series on x_0 by defining a new function $g(x) = f(x + x_0)$ then $g(0) = f(x_0)$ and $g'(x) = f'(x + x_0)$. Repeatedly differentiating, $g^{(n)}(x) = f^{(n)}(x + x_0)$ so we build the Maclaurin series for g , which gives us

$$f(x + x_0) = g(x) = \sum_{n=0}^{\infty} g^{(n)}(0)\frac{x^n}{n!} = \sum_{n=0}^{\infty} f^{(n)}(x_0)\frac{x^n}{n!} \quad (26)$$

or,

$$f(x) = g(x - x_0) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0)\frac{(x - x_0)^2}{2!} + \dots \quad (27)$$

This form is called the Taylor series expansion of f .

4.3 Taylor's Theorem

Lemma 4.3.1: If a real-valued function f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof: Not proved here. □

Theorem 4.3.1: Suppose a function f is n times differentiable on $[a, x]$, then there exists some $\xi \in (a, x)$ such that

$$f(x) = f(a) + (x-a)f'(a) + \dots + \frac{(x-a)^{(n-1)}}{(n-1)!}f^{(n-1)}(a) + R^n(x) \quad (28)$$

Where the remainder term is equal to $R_n(x) = \frac{(x-a)^n}{n!}f^{(n)}(\xi)$.

Proof: Suppose a function f is n times differentiable on $[a, x]$. Let k be a constant defined so that

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^{(n-1)}}{(n-1)!}f^{(n-1)}(a) + \frac{(x-a)^n}{n!}k \quad (29)$$

We can then define the function

$$F(y) = f(x) - f(y) - (x-y)f'(y) - \dots - \frac{(x-y)^{(n-1)}}{(n-1)!}f^{(n-1)}(y) - \frac{(x-y)^n}{n!}k \quad (30)$$

Then $F(x) = 0$ and also $F(a) = f(x) - f(x) = 0$ so we can apply Rolle's theorem to F on $[a, x]$, meaning there is some $\xi \in (a, x)$ such that $F'(\xi) = 0$. □

4.4 Fourier Series

Definition 4.4.1: For a function $f(x)$, $-\pi < x < \pi$, the Fourier series is

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad (31)$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \quad (32)$$

Part II

Linear Algebra

5 Introduction to Linear Algebra

5.1 Mathematical Structures

Definition 5.1.1: Abstraction is the process of simplifying a complex problem via removing unnecessary details in order to apply more general problem solving techniques.

Definition 5.1.2: Groups consist of a set G and a binary operation $*$. They abide by the following axioms:

- **Definition 5.1.3:** Closure, when applying an operation between two elements of the group, the result will also be an element of the group.

$$\forall a, b \in G, a * b = c, c \in G \quad (33)$$

- **Definition 5.1.4:** Associativity, for any elements a, b, c in the group, the order in which the operation is performed does not matter, as long as the elements stay in their original order.

$$\forall a, b, c \in G, a * (b * c) = (a * b) * c \quad (34)$$

- **Definition 5.1.5:** Identity, the group contains an element such that when an operation is performed between an element in the group and the identity, the original element is returned.

$$\exists e \in G, \forall a \in G, a * e = e * a = a \quad (35)$$

- **Definition 5.1.6:** Inverse, for every element in the group, there exists an inverse element such that when an operation is performed between them, the identity is returned.

$$\forall a \in G, \exists b \in G, a * b = e \quad (36)$$

Definition 5.1.7: Fields contain a set F with two binary operations (addition and multiplication) and abide by the following axioms:

- Associativity (defined above).
- **Definition 5.1.8:** Commutativity - when applying an operation between two elements of the field, the order the elements appear in does not matter.

$$\forall a, b \in F, a + b = b + a \text{ and } a \times b = b \times a \quad (37)$$

- **Definition 5.1.9:** Identity, there exists two distinct elements, 0 and 1, in F such that,

$$\exists 0, 1 \in F, \forall a \in F, a + 0 = a \text{ and } a \times 1 = a \quad (38)$$

- **Definition 5.1.10:** Additive inverses, for every element in the field, there exists some other element, called the additive inverse, such that,

$$\forall a \in F, \exists b \in F, a + b = 0 \quad (39)$$

- **Definition 5.1.11:** Multiplicative inverses, for every element in the field, there exists some other element, called the multiplicative inverse, such that,

$$\forall a \in F, \exists b \in F, a \times b = 1 \quad (40)$$

- **Definition 5.1.12:** Distributivity of multiplication over addition. This axiom states that,

$$\forall a, b, c \in F, a \times (b + c) = (a \times b) + (a \times c) \quad (41)$$

Definition 5.1.13: A vector space is a mathematical structure with two sets, scalars F and vectors V , where V forms a commutative group under addition and F forms a field. They abide by the following axioms:

- Associativity (defined above).
- Identity (defined above).
- Distributivity of multiplication over addition (defined above).

5.2 Linear Combinations and Maps

Definition 5.2.1: Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and scalars $a_1, a_2, \dots, a_n$, we can form a linear combination.

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \sum_{i=1}^n a_i\mathbf{v}_i \quad (42)$$

Definition 5.2.2: The set of all linear combinations of a set of vectors is called its span, denoted $\text{span}(V)$.

Definition 5.2.3: A set of vectors S is a spanning set for a vector space V if $\text{span}(S) = V$. We can equivalently say that S spans V .

Definition 5.2.4: A set of vectors S is linearly dependent if there exists a linear combination such that

$$\sum_{i=1}^n a_i\mathbf{s}_i = 0 \text{ and } \exists i, a_i \neq 0 \quad (43)$$

Otherwise, the set is described as linearly independent.

Definition 5.2.5: A subset S of the vector space V is a basis for V if S is a spanning set and is linearly independent.

Theorem 5.2.1: For a vector space V , if S is a basis for V , then it must also be a minimal spanning set.

Proof: Let S be a basis of V . Assume that S is not a minimal spanning set, then there must be some element $s \in S$ such that $\text{span}(S \setminus \{s\}) = V$. Then, s can be written as a linear combination of elements from $S \setminus \{s\}$. That is, we have that

$$1 \cdot \mathbf{s} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \dots + a_n\mathbf{s}_n) = 0 \quad (44)$$

This means that S would be linearly dependent, contradicting the definition of S as a basis. Therefore, S must be a minimal spanning set. □

Definition 5.2.6: The dimension of a vector space V is the cardinality of its basis S .

Definition 5.2.7: If we have a function $f : V \rightarrow W$ where V and W are vector spaces over a field F , then f is a linear map if,

$$\exists \mathbf{u}, \mathbf{v} \in V, \exists a \in F, f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \text{ and } f(a \cdot \mathbf{v}) = a \cdot f(\mathbf{v}) \quad (45)$$

These can also be called linear transformations or vector space homomorphisms.

Definition 5.2.8: If $V = W$ for a linear map f , then f is also a endomorphism.

6 Matrices

6.1 Matrix Operations

When the determinant of a matrix is zero, its columns are linearly dependent vectors. Therefore, if two columns are the same, then the determinant will be zero. Similarly, if a matrix A is identical to another matrix B , except for two columns being swapped, then $\det(A) = -\det(B)$.

Definition 6.1.1: To calculate a determinant, we use the Laplace expansion,

$$\det(C) = \sum_{j=1}^n (-1)^{i+j} c_{(i,j)} M_{(i,j)} \quad 0 < i \leq n, i \in \mathbb{N} \quad (46)$$

where $c_{(i,j)}$ is the coefficient of the term at (i, j) and $M_{(i,j)}$ is the minor at coordinates (i, j) .

Definition 6.1.2: Under a linear map $f : X \rightarrow Y$, $f(a\mathbf{x} + b\mathbf{y}) = af(\mathbf{x}) + bf(\mathbf{y})$, $f(X)$ is known as the column space and is the span of the columns. Its dimension is known as the column rank (this is also the number of linearly independent columns). Row space and rank can be defined similarly.

Definition 6.1.3: The rank of a matrix is its row/column rank after the matrix is put into RREF.

Definition 6.1.4: If $f : X \rightarrow Y$ is a linear map, then the kernel or null space of f is the set of vectors that map to zero. The dimension of the kernel is known as the nullity. The kernel forms a vector space where all linear combinations of kernel elements are in the kernel.

Definition 6.1.5: Given a linear map on vector spaces $T : V \rightarrow W$, if V has finite dimensions then $\text{Rank}(T) + \text{Nullity}(T) = \dim(V)$.

6.2 Systems of Equations and Gaussian Elimination

Definition 6.2.1: A finite set of linear equations is called a system of equations or a linear system.

- **Definition 6.2.2:** A solution is a sequence of numbers $s_1, s_2, \dots, s_n$ such that the assignment $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ satisfies every equation.
- **Definition 6.2.3:** A system is consistent if it has at least one solution.
- **Definition 6.2.4:** A system $A\mathbf{x} = \mathbf{b}$ is homogenous if $\mathbf{b}$ is a column of zeros. These systems have a trivial solution of $\forall x \in \mathbf{x}, x = 0$.

Definition 6.2.5: An augmented matrix is in row echelon form if the first non-zero number in each row is a one, any rows containing only zeros are grouped at the bottom, and if two successive rows are not all zeroes, then the leading one of the higher row is to the left of the leading one in the lower row.

Definition 6.2.6: An augmented matrix is in reduced row echelon form if it is in row echelon form and each column containing a leading one has zeroes everywhere else.

6.3 Geometry

Definition 6.3.1: Given a vector space X , a function $d : X \rightarrow \mathbb{R}$ is a norm if it satisfies:

- **Definition 6.3.2:** Triangular Inequality: $\forall x, y \in X, d(x + y) \leq d(x) + d(y)$.
- **Definition 6.3.3:** Absolute Homogeneity: $\forall x \in X, s \in \mathbb{R}, d(s \cdot x) = |s| \cdot d(x)$.
- **Definition 6.3.4:** Positive Definite: If $d(x) = 0$ then $x = 0$.

Definition 6.3.5: The Euclidean norm is the length of a vector and is given for $\mathbf{x} \in \mathbb{R}^n$ to be

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n (x_i)^2} \quad (47)$$

Any non-unit vector $\mathbf{x}$ can be used to define a unit vector $\hat{\mathbf{x}}$ in the direction of $\mathbf{x}$ by using

$$\hat{\mathbf{x}} = \frac{1}{\|\mathbf{x}\|} \mathbf{x} \quad (48)$$

Definition 6.3.6: The Manhattan distance is given by

$$d_M(\mathbf{x}) = \sum_{i=1}^n |x_i| \quad (49)$$

Definition 6.3.7: Given $p \in \mathbb{R}^+$, we can define the p -norm $\|\mathbf{x}\|_p$ on $\mathbb{R}^n$ as

$$\|\mathbf{x}\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad (50)$$

Using limits on the p -norm, we get

$$\lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \max(\mathbf{x}) \quad \lim_{p \rightarrow 0} \|\mathbf{x}\|_p = \infty \quad (51)$$

Definition 6.3.8: Two vectors are orthogonal if their dot product is zero.

Definition 6.3.9: Two vectors are normal if their Manhattan norm is one.

Definition 6.3.10: A set of vectors is orthonormal if they are all both orthogonal and normal.

7 Eigenvectors and Eigenvalues

7.1 Introduction to Eigenvectors and Eigenvalues

Definition 7.1.1: Let A be an $n \times n$ matrix. A non-zero vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq 0$ is an eigenvector of A if, for some scalar λ ,

$$A\mathbf{x} = \lambda\mathbf{x} \quad (52)$$

Where λ is the eigenvalue of A corresponding to the eigenvector $\mathbf{x}$.

Definition 7.1.2: The equation $\det(\lambda I - A) = 0$ is called the characteristic equation of A . In general, this is a polynomial where n is in the order of A .

$$p(\lambda) = \lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \dots + c_n \quad (53)$$

Theorem 7.1.1: If A is an $n \times n$ matrix, then λ is an eigenvalue of A iff it satisfies $\det(\lambda I - A) = 0$.

Proof: By definition, λ is an eigenvalue of A iff $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq 0$. We have

$$A\mathbf{x} = \lambda\mathbf{x} \iff A\mathbf{x} = \lambda I\mathbf{x} \iff (\lambda I - A)\mathbf{x} = 0 \quad (54)$$

By theorem about invertible matrices, the last equation has solution $\mathbf{x} \neq 0$ iff $\det(\lambda I - A) = 0$.

□

Definition 7.1.3: For a matrix A with eigenvalue λ , the null space of $\lambda I - A$ is called its eigenspace where the non-zero vectors in this space are the eigenvectors corresponding to λ .

Let λ_0 be an eigenvalue of a matrix A :

- **Definition 7.1.4:** The algebraic multiplicity of λ_0 is the power k with which $(\lambda - \lambda_0)$ appears as a factor of $\det(\lambda I - A)$.
- **Definition 7.1.5:** The geometric multiplicity of λ_0 is the dimension of the eigenspace corresponding to λ_0 .

Definition 7.1.6: Principle component analysis is the process of identifying the principle components (most important eigenvectors) of a data set. It takes a data set in n dimensions and outputs an $n \times n$ matrix, demonstrating how each dimension varies with each other dimension (i.e., it captures the variance of one dimension of the data with another):

1. Pretend we have some covariance matrix $COV_{n \times n}$.
2. Find the n eigenvalues and corresponding eigenvectors of $COV_{n \times n}$.
3. Order the eigenvalues in decreasing order to get the importance of each.
4. Work out what percentage of the total value each eigenvalue contributes and take the top k values, contributing to 80% of the total.
5. Let $P_{n \times k}$ be a matrix with the k chosen eigenvectors as column vectors.

7.2 Complex Matrices

For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\overline{\overline{\mathbf{u}}} = \mathbf{u}$.
- $\overline{k\mathbf{u}} = \overline{k}\overline{\mathbf{u}}$.
- $\overline{\mathbf{u} + \mathbf{v}} = \overline{\mathbf{u}} + \overline{\mathbf{v}}$.
- $\overline{\mathbf{u} - \mathbf{v}} = \overline{\mathbf{u}} - \overline{\mathbf{v}}$.

If A is an $m \times k$ complex matrix and B is a $k \times n$ matrix, then:

- $\overline{\overline{A}} = A$.
- $\overline{A^T} = (\overline{A})^T$.
- $\overline{AB} = \overline{A}\overline{B}$.

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\mathbf{u} \cdot \mathbf{v} = \overline{\mathbf{v} \cdot \mathbf{u}}$.
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
- $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ and $\mathbf{u} \cdot (k\mathbf{v}) = \overline{k}(\mathbf{u} \cdot \mathbf{v})$.
- $\mathbf{v} \cdot \mathbf{v} \geq 0$ and $\mathbf{v} \cdot \mathbf{v} = 0$ iff $\mathbf{v} = \mathbf{0}$.

Theorem 7.2.1: If λ is an eigenvalue of a real $n \times n$ matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $\overline{\lambda}$ is also an eigenvalue of A and $\overline{\mathbf{x}}$ is a corresponding eigenvector.

Proof: Since A is real, we have $A\overline{\mathbf{x}} = \overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$.

□

Theorem 7.2.2: If A is a real symmetric matrix, then all (complex) eigenvalues of A are real.

Proof: Let $\lambda \in \mathbb{C}$ be an eigenvalue of A and $\mathbf{x} \in \mathbb{C}^n$ a corresponding eigenvector. Take the complex conjugate of both sides of the equation $A\mathbf{x} = \lambda\mathbf{x}$. We get $\overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}}$, and, since $A = \overline{A}$, it follows that $A\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$. Then, using $A = A^T$, we compute the number $\overline{\mathbf{x}}^T A \mathbf{x}$ in two different ways:

$$\overline{\mathbf{x}}^T A \mathbf{x} = \overline{\mathbf{x}}^T (A\mathbf{x}) = \overline{\mathbf{x}}^T (\lambda\mathbf{x}) = \lambda(\overline{\mathbf{x}}^T \mathbf{x}) = \lambda(\mathbf{x} \cdot \mathbf{x}) = \lambda\|\mathbf{x}\|^2 \quad (55)$$

$$\overline{\mathbf{x}}^T A \mathbf{x} = (\overline{A\mathbf{x}})^T \mathbf{x} = (\overline{\lambda\mathbf{x}})^T \mathbf{x} = \overline{\lambda}(\overline{\mathbf{x}}^T \mathbf{x}) = \overline{\lambda}(\mathbf{x} \cdot \mathbf{x}) = \overline{\lambda}\|\mathbf{x}\|^2 \quad (56)$$

Since $\mathbf{x} \neq \mathbf{0}$, we have $\|\mathbf{x}\| \neq 0$, so $\lambda\|\mathbf{x}\|^2 = \overline{\lambda}\|\mathbf{x}\|^2$ implies $\lambda = \overline{\lambda}$, meaning $\lambda \in \mathbb{R}$.

□

7.3 Inner Product Spaces

7.4 The Gram-Schmidt Process

8 Decomposition

8.1 LU Decomposition

Definition 8.1.1: Matrix decomposition or factorisation is the process of representing a matrix as a product of two or more matrices with useful properties.

Definition 8.1.2: An elementary matrix is a matrix which differs from the identity matrix as a result of one elementary row operation (switching, addition, or multiplication).

Definition 8.1.3: LU decomposition is a method for solving linear systems by reducing the problem into two triangular systems. This method can be used on a square matrix A if U is its row echelon form, which was produced without any row exchanges being performed. The method is as follows:

1. Keep track of the row operations used to compute U by Gaussian elimination.
2. Let $E_1, \dots, E_k$ be the corresponding elementary matrices for each row operation.
3. Then the inverse matrices $(E_1)^{-1}, \dots, (E_k)^{-1}$ are easy to find.
4. Compute $L = (E_1)^{-1}(E_2)^{-1} \dots (E_k)^{-1}I$ by applying the corresponding row operations to the identity matrix.

Definition 8.1.4: A permutation matrix is a square matrix P obtained from I by permuting its rows.

Definition 8.1.5: A PLU decomposition of a square matrix A is a representation $A = PLU$ where P is a permutation matrix, L is lower triangular, and U is upper triangular.

8.2 Eigendecomposition

Definition 8.2.1: Eigendecomposition is the process of decomposing a matrix into PDP^{-1} form where P are eigenvectors and D is a diagonal matrix of eigenvalues.

Definition 8.2.2: Square matrices A and B are called similar if $A = P^{-1}BP$ for some invertible P . These matrices will have the same determinant, rank, trace, nullity, eigenvalues, etc.

Theorem 8.2.1: Matrices A and B are similar iff they represent the same linear operator $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, possibly in different bases.

Proof: Proof omitted.

□

Definition 8.2.3: A matrix A is called diagonalisable if it is similar to a diagonal matrix, i.e., there exists an invertible matrix P such that $P^{-1}AP$ is diagonal. Then P is said to diagonalise A .

Theorem 8.2.2: An $n \times n$ matrix is diagonalisable iff it has n linearly independent eigenvectors.

Proof: ($\implies$). Assume that there is an invertible matrix P and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that $D = P^{-1}AP$, or $AP = PD$. Denote the i column vectors of P as $\mathbf{p}_i$, then

$$AP = A[\mathbf{p}_1 | \dots | \mathbf{p}_n] = [A\mathbf{p}_1 | \dots | A\mathbf{p}_n] \quad (57)$$

On the other hand,

$$PD = [\lambda_1 \mathbf{p}_1 | \dots | \lambda_n \mathbf{p}_n] \quad (58)$$

Since $AP = PD$, we conclude that $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$ for all $1 \leq i \leq n$. Since P is invertible, its rank is n and so the vectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ are linearly independent, making each of them an eigenvector.

□

($\impliedby$).

□

8.3 QR Decomposition

8.4 Linear Regression

8.5 Spectral Decomposition

8.6 Applications of Linear Algebra

Part III

References

9 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

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10 Cited Works